

Question number	Answer	Mark
2(a)(i)	The only correct answer is B <i>A is incorrect as the arrow labelled Q is the activation energy</i> <i>C is incorrect as the arrow labelled Q is the activation energy</i> <i>D is incorrect as the arrow labelled Q is the activation energy</i>	(1)

Question number	Answer	Mark
2(a)(ii)	The only correct answer is C <i>A is incorrect as monophenol is the reactant and is located at Y</i> <i>B is incorrect as monophenol is the reactant and is located at Y</i> <i>D is incorrect as monophenol is the reactant and is located at Y</i>	(1)

Question number	Answer	Mark
2(a)(iii)	The only correct answer is C <i>A is incorrect as P will stay the same and Q will get shorter as the activation energy decreases</i> <i>B is incorrect as P will stay the same and Q will get shorter as the activation energy decreases</i> <i>D is incorrect as P will stay the same and Q will get shorter as the activation energy decreases</i>	(1)

Question number	Answer	Additional guidance	Mark
2(b)(i)	- tangent drawn at 3 minutes on the outside of the curve (1) - value between 6.5 and 12.6 to max one decimal place (1)	ACCEPT with or without a minus sign Bald correct answer = 2 marks (answer / bald answer} in range but with too many decimal places = 1 mark	(2)

Question number	Answer	Additional guidance	Mark
2(b)(ii)	An explanation that includes two of the following points: • {curve / gradient / graph / fall} will be steeper and {plateau / reach zero sooner} (1) • (optimum temperature results in) more {energy / collisions / enzyme-substrate complexes} (1) • (levels off sooner) substrate will run out (1)		(2)